

Bias Variance Analysis

Ref: Domingos, Pedro. "A unified bias-variance decomposition." *Proceedings of 17th international conference on machine learning*. Morgan Kaufmann, 2000.

For Regression

Error

Error of a learning method can be decomposed into *bias* and *variance*.

Bias: Deviation from the true value (inherent weakness of the learning method)

Variance: Variance of the prediction over different training sets (variation in the prediction because the training set is varied).

Error

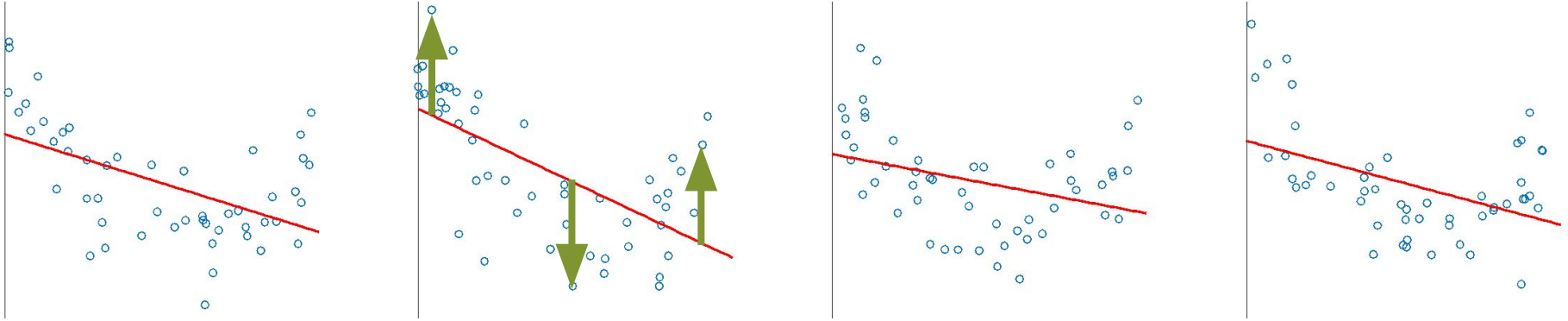
Error depends on the learning method (f) and on the training set (D_i).

For the given test example X , the prediction is $f(X)$.

Actually since this depends on the training set D_i , we can write $f(X; D_i)$

Bias

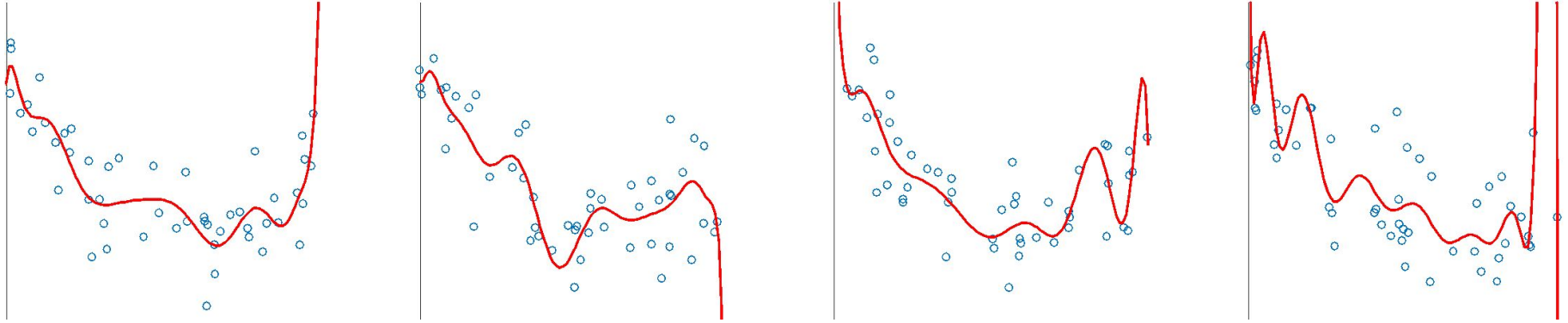
Actual relationship may be quadratic. But we are trying to fit a line.



Regardless of training sample, or size of training sample, model will produce consistent errors

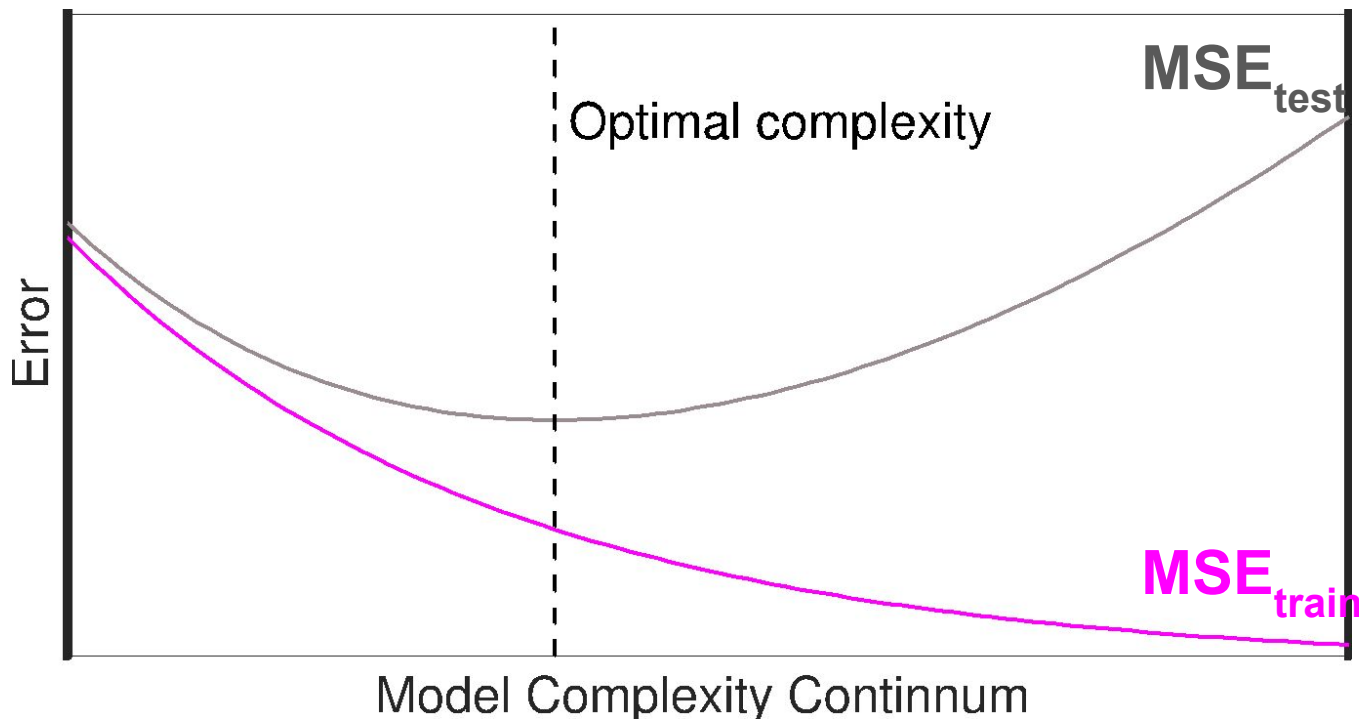
Variance

We are trying to fit degree 8 polynomial



Different samples of training data yield different model fits

Bias-Variance Trade Off Is Revealed Via Test Set Not Training Set



Bias increases $\longleftrightarrow$ $\longrightarrow$ **Variance increases**

Training sets

Let $D = \{D_1, \dots, D_i, \dots, D_{10}\}$, i.e., we are having 10 different training sets drawn from the same distribution.

Size of each D_i is the same. Let us say $|D_i| = n$.

Each training set will be like

$$D_i = \{(X_1, t_1), \dots, (X_n, t_n)\}$$

Note, t is the target and $y = f(X; D_i)$ is the prediction

Mean prediction for x is called y_m

- $y_m = \text{Mean} \{f(X; D_1), f(X; D_2), \dots, f(X; D_{10})\}$
- This is nothing but average prediction over the training sets.

Square Loss

$$L(y_i, y_j) = (y_i - y_j)^2$$

In regression we usually use the square loss.

Bias

- On average, deviation from the true prediction.
- For X , bias in the prediction is, $B(X) = L(E_t t, y_m) = (E_t t - y_m)^2$
$$= \left(E_t t - \frac{f(X; D_1) + \dots + f(X; D_{10})}{10} \right)^2$$

Variance

- Variance in the prediction
- For X , variance in the prediction is, $V(X) = \frac{L(f(X; D_1), y_m) + \dots + L(f(X; D_{10}), y_m)}{10}$

Note, for a single example (X,y) these are bias and variance (across the training sets)

Bias and Variance

- Bias and variance has to be found by averaging over the entire feature-space.
 - Bias = $E_X[B(X)]$
 - Variance = $E_X[V(X)]$
-
- In practice, we take average over the Test Set.

Test Set

- Let D_s be the test set.
- Let $D_s = \{(X_1, t_1), \dots, (X_s, t_s)\}$
- Let $|D_s| = s$

- Bias = $\frac{1}{s} \sum_{k=1}^s B(X_k)$
- Variance = $\frac{1}{s} \sum_{k=1}^s V(X_k)$
- Note, this is the average Bias and Variance over the Test Set.
- The summation is over all Test Examples.

Formal Derivation

Formally (we introduce t ...)

- Let D be the set of training sets (each of same size)
- Let t be the target (true value) for the given x .
- Note, t is a random variable. It depends on x . But is independent of any training set.
- y is the prediction which depends on the training set, hence we write $y = f(x, D_i)$ where D_i is the training set. y depends on D_i but is independent of t .

For the given x , let t be the target, and D be the set of training sets

- The expected loss incurred for the prediction of x is

$$\begin{aligned} E_{D,t}[L(t,y)] &= E_{D,t}[(t-y)^2] \\ &= E_{D,t}(t - E_t t + E_t t - y)^2 \\ &= E_t(t - E_t t)^2 + E_{D,t}(E_t t - y)^2 \\ &= E_t(t - E_t t)^2 + E_{D,t}(E_t t - E_D y + E_D y - y)^2 \\ &= E_t(t - E_t t)^2 + (E_t t - E_D y)^2 + E_D(E_D y - y)^2 \\ &= N(x) + B(x) + V(x) \end{aligned}$$

Here $N(x)$ is the noise, $B(x)$ is the Bias and $V(x)$ is the variance for x .

Note: $E_D y = y_m$

For Classification

0-1 Loss

$$L(y_i, y_j) = \begin{cases} 0, & \text{if } y_i = y_j \\ 1, & \text{otherwise} \end{cases}$$

In classification we usually use 0-1 Loss

Main prediction for X is called y_m

- $y_m = \text{Mode} \{f(X; D_1), f(X; D_2), \dots, f(X, D_{10})\}$
- This is nothing but majority (most frequent) prediction of f over the training sets.
 - In case of a Tie we break it randomly

The Bayes prediction for X

- y^* is the Bayes prediction for X
- Since we know the distributions, this can be found from the Bayes Classifier

Bias

- Deviation from the Bayes classifier.
- For X , bias in the prediction is, $B(X) = L(y_m, y^*)$

Variance

- Variance in the prediction
- For X , variance in the prediction is, $V(X) = \frac{L(f(X; D_1), y_m) + \dots + L(f(X; D_{10}), y_m)}{10}$

Note, y_m , y^* are the main prediction and the Bayes prediction for the x , respectively.

The decomposition for 0-1 loss

$$\text{Noise} = N(x) = E_t[L(t, y^*)]$$

$$\text{Error for } x, E_{D,t}[L(t, y)] = c_1 E_t[L(t, y^*)] + L(y^*, y_m) + c_2 E_D[L(y_m, y)]$$

$$= c_1 N(x) + B(x) + c_2 V(x)$$

c_1 and c_2 are complicated expressions whose values are beyond the scope of this class.

Note

- We have taken 10 different training sets.
- In general, one may consider different number of training sets to estimate Bias and Variance in practice. In theory we take expectation over all training sets of the given size.

In case of classification, the Bayes error forms the noise. One may refer to the paper

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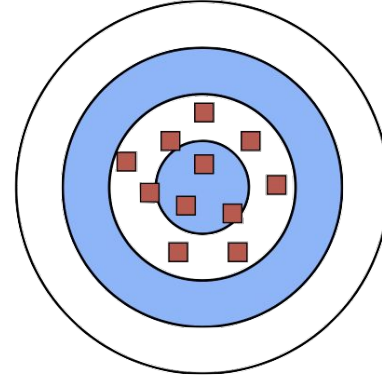
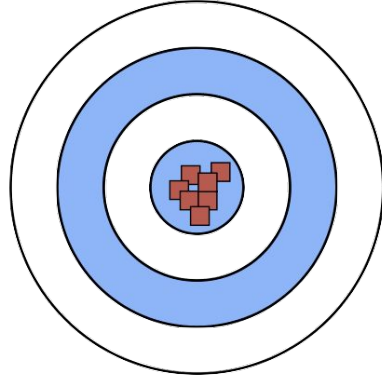
For formal derivations, refer to

- **Ref:** Domingos, Pedro. "A unified bias-variance decomposition." *Proceedings of 17th international conference on machine learning*. Morgan Kaufmann, 2000.

Low Variance
(Precise)

High Variance
(Not Precise)

Low Bias
(Accurate)



High Bias
(Not Accurate)

